Layer 1: Sample Collection and Pre-Analytical Modelling

Swab Absorption, Elution Efficiency, Inhibitor Co-Elution, Transit Degradation, and the Wound-to-Chamber Dilution Chain

Eton iGEM 2026, Dry Lab | Layer 1 of 17-Layer Biosensor Pipeline | `preanalytical_model.py` · `uncertainty_propagation.py`

Layer 1 is the pre-analytical layer of the device model chain: it converts the in-situ wound-fluid enzyme concentration $[E]_{wound}$ predicted by Layer 0 into the effective enzyme concentration actually incubating on the electrode, $[E]_{chamber}$ – the true input to Layer 3's bulk-transport PDE and, downstream, to Layer 9's surface kinetics. Every physical step between swab and cartridge (absorption, elution, buffer mixing, transit decay, filtration, chamber loading) either dilutes or degrades the analyte, and every one of Layer 9/13/15's calibration and clinical numbers is only as good as this layer's account of that dilution.

Mass-balance correction. contents.md Sec. 1.2 states the eluate concentration as $[E]_{eluate} = [E]_{wound}\,\eta_{elut}$, i.e. elution efficiency alone. This omits the dilution of the absorbed wound fluid into the (much larger) elution buffer volume: a swab absorbing V_{abs} of wound fluid and releasing a fraction η_{elut} of its analyte into $V_{buffer} = 200\,\mu$L does not return the eluate at the wound-fluid concentration scaled only by efficiency – mass balance requires an additional factor V_{abs}/V_{buffer} (Sec. II). Because $V_{abs}/V_{buffer} \approx 0.1$–$1.0$ over the literature V_{abs} range, this is not a small correction: it is typically the single largest dilution step in the whole pre-analytical chain (Sec. VIII), and it is not present anywhere in Layers 0, 9, 13 or 15's own treatment of $[E]_t$ (Sec. IX). This report uses the corrected form throughout and flags the original formula as superseded.

I. Swab Absorption Volume

Distribution and parameters. contents.md Sec. 1.1 quotes a literature range $V_{abs} \approx 20$–$200\,\mu$L per swab pass on a moderate-exudate wound, and specifies a log-normal model. No single wet-lab source for this range was confirmed for this project's specific swab material during this pass (the "key literature to find" list in contents.md – Vowden, Gardner – was not located in the time available; see Limitations). Treating 20–200 μL as the central 90% interval (5th–95th percentile) of a log-normal distribution gives

$$V_{abs} \sim \text{LogNormal}(\mu, \sigma^2), \quad \mu = \ln\sqrt{20 \times 200} = 4.146, \quad \sigma = \frac{\ln(200/20)}{2 \times 1.645} = 0.700 \tag{I.1}$$

so the geometric-mean central estimate is $\sqrt{20 \times 200} = 63.2\,\mu$L, with a geometric standard deviation of $e^{0.700} = 2.01\times$ (i.e. the 5th/95th percentiles sit at $63.2/2.01^{1.645}$ and $63.2 \times 2.01^{1.645}$, reproducing 20 and 200 μL). The resulting coefficient of variation, $\sqrt{e^{\sigma^2} - 1} = 79.5\%$, already exceeds the $\sim$50%

placeholder contents.md Sec. 1.7 anticipates for this term – Sec. VIII confirms V_{abs} is in fact the dominant uncertainty source, but by a wider margin than the qualitative estimate suggested.

Drivers not modelled quantitatively here. Exudate level (low/moderate/heavy), swab material (cotton, foam, polyester) and contact time/pressure all shift the location of this distribution; contents.md gives only the pooled moderate-exudate range, so this report treats V_{abs} as a single random variable rather than a function of these covariates. Resolving this decomposition is listed as a priority wet-lab measurement (Limitations, item 1).

II. Elution Efficiency and the Corrected Eluate Concentration

Size-dependent bands. contents.md Sec. 1.2 gives elution efficiency η_{elut} in three molecular-weight bands: small proteins (< 10 kDa) 0.7–0.9; medium proteins (30–100 kDa, covering both HNE at 30 kDa and MMP-9 at 92 kDa) 0.4–0.7; large proteins/complexes 0.2–0.5. These are unconfirmed placeholders (no swab-elution-specific source was located for either target enzyme; see Limitations). This report uses the band midpoint, $\eta_{elut} = 0.55$, as the central value for both MMP-9 and HNE.

The missing dilution term. As stated in the introduction, contents.md's eluate formula omits the volume ratio between the absorbed wound fluid and the elution buffer it is released into. By mass balance, moles of analyte released $= \eta_{elut}\,[E]_{wound}\,V_{abs}$, and (taking the total eluate volume as $\approx V_{buffer}$, since $V_{buffer} \gg$ the small retained volume) the corrected eluate concentration is

$$[E]_{eluate} \;=\; [E]_{wound}\,\eta_{elut}\,\frac{V_{abs}}{V_{buffer}} \tag{II.1}$$

At the central values ($V_{abs} = 63.2\,\mu$L, $V_{buffer} = 200\,\mu$L, $\eta_{elut} = 0.55$), the additional factor $V_{abs}/V_{buffer} = 0.316$ multiplies directly onto the elution efficiency, so the effective wound-to-eluate recovery is $0.55 \times 0.316 = 17.4\%$, not 55% as Sec. 1.2's uncorrected formula would imply. Equation (II.1) is used in place of Sec. 1.2's formula for every downstream calculation in this report (Sec. VII–IX) and should be back-ported into contents.md and into any code already written against the original expression.

Material dependence – a caution beyond the MW-band model. contents.md's bands are keyed only to molecular weight, but elution efficiency is also a strong function of swab/dressing *material*, and the effect can be much larger than the MW bands suggest. Wiegand & Hipler (2013) [5] showed that a superabsorbent polyacrylate wound dressing binds MMP-2 and MMP-9 so strongly that a subsequent PBS elution step released essentially none of the bound MMP-2 and only marginal MMP-9 – i.e. $\eta_{elut} \to 0$ for that material, independent of molecular weight. Singh et al. (2022) [6] similarly found most commercial superabsorbent dressings fully sequester MMP-2/9 over multi-day contact. Cotton, foam and polyester swabs (the materials contents.md Sec. 1.1 specifies) are chemically distinct from polyacrylate superabsorbers and are not expected to bind MMP-9 this strongly, but no confirmatory elution data exists for the fabricated device's actual swab material. If the chosen swab material has any polyacrylate/superabsorbent component (common in high-capacity wound swabs), η_{elut} for MMP-9 could sit far below the 0.4–0.7 band used here – this is flagged as a priority swab-material screening measurement (Limitations, item 2).

III. Inhibitor Co-Elution and the Enzyme/Inhibitor Ratio Shift

TIMP-1 (28 kDa), A1AT (52 kDa) and SLPI (11.7 kDa) co-elute with the target enzymes and continue to inhibit them inside the cartridge during incubation (contents.md Sec. 1.3; the resulting in-device binding kinetics are modelled in Layer 3). Applying Eq. (II.1) to each inhibitor with its own size-dependent $\eta_{elut,I}$ (SLPI 0.8, TIMP-1 0.6, A1AT 0.5, per contents.md Sec. 1.3) gives

$$[I]_{chamber} = [I]_{wound}\,\eta_{elut,I}\,\frac{V_{abs}}{V_{buffer}}\,D_f \tag{III.1}$$

the same chain as the target enzyme (Sec. VII), with $\eta_{elut,I}$ in place of $\eta_{elut,E}$. Because V_{abs}/V_{buffer}, transit decay and D_f are common to both species and cancel in a ratio, the *shift* in the inhibitor/enzyme ratio between wound fluid and chamber depends only on the relative elution efficiencies:

$$\text{enrichment} = \frac{[I]_{chamber}/[E]_{chamber}}{[I]_{wound}/[E]_{wound}} = \frac{\eta_{elut,I}}{\eta_{elut,E}} \tag{III.2}$$

Fig. 1 (left) shows this computed for the three pairs contents.md flags: SLPI is enriched 1.45× relative to HNE (small, high-efficiency SLPI outpaces medium-efficiency HNE, exactly the direction contents.md Sec. 1.3 anticipates), TIMP-1 is enriched 1.09× relative to MMP-9 (both sit in the same medium/large band, so the shift is small), and A1AT is very slightly *depleted* (0.91×) relative to HNE. The practical implication is that any wound-fluid-calibrated in-device inhibition model (Layer 3) that assumes the wound-fluid inhibitor/enzyme ratio carries through unchanged will under-correct the SLPI-HNE inhibition by ∼45% and should use the chamber-corrected ratios instead.

IV. Elution Buffer Effects on Matrix: pH, Ionic Strength, Ca^{2+}, and Redox Species

Ionic strength and Ca^{2+}. With PBS (150 mM NaCl, 0.9 mM Ca^{2+}) as elution buffer and wound-fluid Ca^{2+} at ∼1–2 mM (contents.md Sec. 1.4; consistent with the ∼1–2 mM estimate used for the Edinburgh Ca^{2+} artefact in Layer 7), the PBS-dominated eluate keeps ionic strength close to 150 mM and net [Ca^{2+}] in the 1–2 mM range – simplifying Layer 7's Ca^{2+}-artefact model, since [Ca^{2+}] entering the chamber is approximately known rather than needing wound-specific measurement.

pH: exact mixing, not complete buffering. contents.md Sec. 1.4 cautions that complete buffering to pH 7.4 only holds if $V_{wound} \ll V_{buffer}$; otherwise the mixed pH must be computed from proton mass balance,

$$\mathrm{pH}_{chamber} = -\log_{10}\left(\frac{V_{wound}\,10^{-\mathrm{pH}_{wound}} + V_{buffer}\,10^{-\mathrm{pH}_{buffer}}}{V_{wound} + V_{buffer}}\right). \tag{IV.1}$$

For the worked example in contents.md ($\mathrm{pH}_{wound} = 6.5$, $V_{wound} = 50\,\mu$L into $V_{buffer} = 200\,\mu$L PBS at pH 7.4), Eq. (IV.1) gives $\mathrm{pH}_{chamber} = 7.02$, confirming the "≈ 7.0" estimate in contents.md to two decimal places. Fig. 1 (right) sweeps this over the full wound-pH range (6.5–8.2, per the

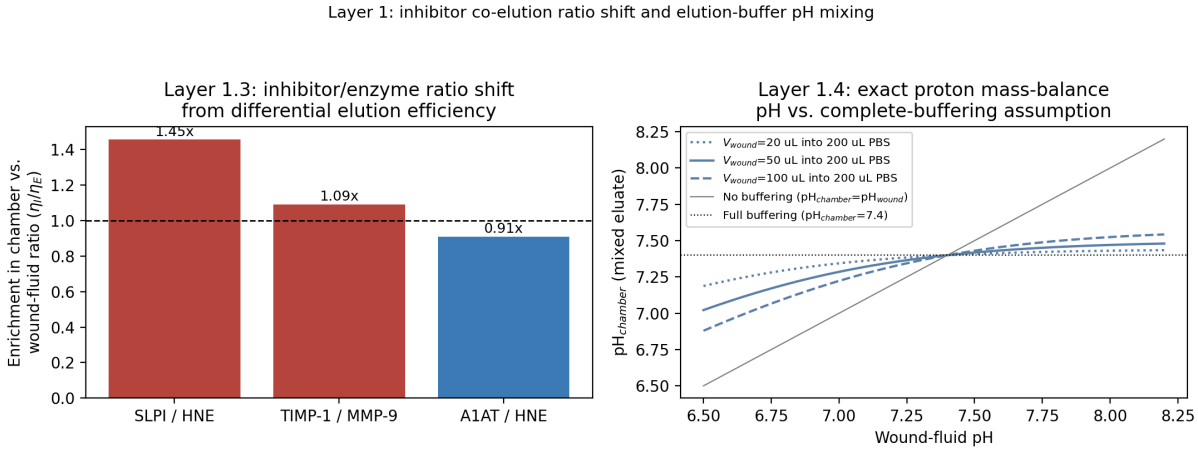


Figure 1: **Layer 1.3/1.4: inhibitor co-elution and pH mixing.** Left: enrichment of each inhibitor relative to its target enzyme in the chamber vs. the wound-fluid ratio, from differential elution efficiency alone (Eq. III.2). Right: exact proton-mass-balance $\mathrm{pH}_{chamber}$ (Eq. IV.1) vs. wound-fluid pH, for three absorbed volumes into a fixed 200 μL PBS buffer, compared against the no-buffering and complete-buffering limits.

README's Layer 0 range) and shows the completely-buffered assumption (pH = 7.4 flat) is a reasonable approximation only once $V_{wound} \lesssim$ 20–50 μL; at the upper end of the V_{abs} distribution (Sec. I, up to 200 μL) with a full 200 μL buffer, the true chamber pH can deviate from 7.4 by 0.2–0.3 units even for physiological wound pH. contents.md notes a 0.4-unit pH shift moves E°_{MB} by 24 mV (comparable to the SWV peak width, Layer 6/10) – so $\mathrm{pH}_{chamber}$ from Eq. (IV.1), not an assumed 7.4, should be propagated to Layers 6, 10 and 12, as contents.md already specifies.

Redox species. Wound-fluid glutathione (∼1 mM), ascorbate (∼0.1 mM) and H_2O_2 (∼10–100 μM) co-elute into the chamber at the same η_{elut}-and-dilution chain as any other small solute (Sec. II), giving $[\mathrm{GSH}]_{chamber}$, $[\mathrm{ascorbate}]_{chamber}$, $[\mathrm{H_2O_2}]_{chamber}$ as direct inputs to Layer 7's SAM-oxidation model and Layer 11's MB redox-interference model, exactly as contents.md specifies; no additional model is developed here beyond noting they follow Eq. (II.1) with their own (unconfirmed, likely small-molecule/high) η_{elut}.

V. Analyte Stability During Transit

Between collection and cartridge loading the eluate sits at ambient temperature for $t_{transit}$ (contents.md: typically 0–15 min in clinic, up to 2 h/120 min in a field setting), decaying by first-order kinetics,

$$[E](t_{transit}) = [E]_{eluate}(0)\,\exp(-k_{deg}\,t_{transit}), \qquad k_{deg} \in [0.001, 0.01]\ \mathrm{min}^{-1} \tag{V.1}$$

driven by MMP-9 auto-proteolysis, ongoing endogenous inhibitor binding, and surface adsorption to the swab or tube (contents.md Sec. 1.5). Fig. 3 (right) sweeps Eq. (V.1) across the full k_{deg} range and both transit regimes: at the contents.md-quoted 15-min example and the range midpoint $k_{deg} = 0.0055\,\mathrm{min}^{-1}$, survival is $\exp(-0.0055 \times 15) = 92.1\%$ (matching the "≈93%, 7%

loss" figure in contents.md at $k_{deg} = 0.005$); at the field-worst-case 120 min and the upper-bound $k_{deg} = 0.01\,\mathrm{min}^{-1}$, survival falls to 30.1% – a loss contents.md's 15-minute worked example does not surface, and one that matters operationally if the device is used outside a clinic with immediate processing.

VI. Particulate Filtration and Inlet Pressure Drop

Fibrin clots (∼1–10 μm), bacterial aggregates (∼1–5 μm) and cellular debris (∼5–20 μm) require an inlet filter of pore size $d_{pore} \in [0.45, 2]\,\mu$m (contents.md Sec. 1.6 [first instance]) to avoid electrode fouling and inlet blockage (Layer 7 fouling model). The Hagen–Poiseuille pressure drop across a parallel-pore membrane of this type is

$$\Delta P_{filter} = \frac{8\,\eta\,L\,Q}{\pi\,r_{pore}^{4}\,N_{pores}} \tag{VI.1}$$

which subtracts directly from the capillary driving pressure available for chamber fill (Layer 2). Using representative values pending Layer 2's actual fill-time/geometry output – wound-fluid viscosity $\eta = 1.5 \times 10^{-3}\,\mathrm{Pa\,s}$ (∼1.5× water), membrane thickness $L = 20\,\mu$m, flow rate $Q = 5\,\mu$L/s, mid-range pore radius $r_{pore} = 0.75\,\mu$m, and $N_{pores} = 2 \times 10^{6}$ over a ∼5×5 mm patch – Eq. (VI.1) gives $\Delta P_{filter} \approx 0.60\,$kPa. This is small compared to typical capillary pressures for aqueous fluids on hydrophilic surfaces (∼1–3 kPa for the channel dimensions in Layer 2's own plan), so filtration is unlikely to stall fill at these illustrative parameters, but this has not been checked against Layer 2's actual ΔP_{cap} once that layer's model exists – listed as an open cross-layer check (Limitations, item 5).

VII. Net Chamber Concentration and Composite Recovery Factor

Combining Eq. (II.1), the transit decay of Eq. (V.1), and the cartridge-loading factor $D_f = V_{eluate}/V_{chamber}$ (contents.md Sec. 1.6 [second instance]; worked example $D_f = 30/37.5 = 0.8$) gives the full corrected pre-analytical chain,

$$[E]_{chamber} = [E]_{wound}\,\eta_{elut}\,\frac{V_{abs}}{V_{buffer}}\,\exp(-k_{deg}\,t_{transit})\,D_f \tag{VII.1}$$

superseding contents.md's Sec. 1.6 formula (which used the uncorrected $[E]_{eluate}$ of Sec. 1.2). Defining the composite recovery factor $R_{total} = [E]_{chamber}/[E]_{wound}$, the central-value point estimate at $V_{abs} = 63.2\,\mu$L, $\eta_{elut} = 0.55$, $k_{deg} = 0.0055\,\mathrm{min}^{-1}$, $D_f = 0.8$ is

$$R_{total}(t_{transit}{=}15\,\mathrm{min}) = 12.8\%, \qquad R_{total}(t_{transit}{=}120\,\mathrm{min}) = 7.2\%. \tag{VII.2}$$

Applied to Layer 0's confirmed $t = 30\,$min wound-fluid concentrations (chronic 31.9 nM, healing 6.2 nM; Ladwig 2002 [1], as used identically in Layer 9/13/15), the point-estimate chamber concentrations are:

Wound state	$[E]_{wound}$ (nM)	$[E]_{chamber}$, 15-min transit (nM)	$[E]_{chamber}$, 120-min transit (nM)
Chronic	31.9	4.09	2.29
Healing	6.2	0.79	0.45

i.e. at central-value pre-analytical parameters, the concentration actually reaching the electrode is roughly an order of magnitude below the wound-fluid value Layer 9 currently uses directly as its enzyme input (Sec. IX).

VIII. Monte Carlo Uncertainty Propagation

contents.md Sec. 1.7 proposes a quadrature error-propagation formula,

$$\sigma_{[E]}/[E] = \sqrt{(\sigma_{V_{abs}}/V_{abs})^2 + (\sigma_\eta/\eta)^2 + (\sigma_{k_{deg}}/k_{deg})^2},$$

which assumes small, independent, roughly Gaussian relative errors. Because V_{abs} spans a full decade (Sec. I) this linearisation is a poor approximation; `uncertainty_propagation.py` instead runs a direct Monte Carlo ($N = 3 \times 10^5$ draws) over $V_{abs} \sim$ LogNormal (Eq. I.1), $\eta_{elut} \sim$ Triangular$(0.4, 0.55, 0.7)$, $k_{deg} \sim$ Uniform$(0.001, 0.01)\,\text{min}^{-1}$, $t_{transit} \sim$ Uniform$(0, 15)$ or Uniform$(0, 120)$ min, and $D_f \sim \mathcal{N}(0.8, 0.12)$ (truncated > 0; the $\pm 15\%$ relative spread on D_f is an assumed pipetting/loading tolerance, not itself sourced from contents.md), propagated through Eq. (VII.1).

Scenario	Mean (nM)	Median (nM)	[P5, P95] (nM)	$P([E]_{chamber} < \text{LOD})$
Chronic, 0–15 min transit	5.44	4.18	[1.26, 13.81]	99.2%
Healing, 0–15 min transit	1.06	0.81	[0.25, 2.67]	100.0%
Chronic, 0–120 min transit	4.21	3.16	[0.87, 11.05]	99.6%
Healing, 0–120 min transit	0.82	0.61	[0.17, 2.15]	100.0%

Table 1: Monte Carlo summary of $[E]_{chamber}$ against Layer 15's placeholder-k_{surf} LOD (23.9 nM).

The composite recovery factor itself is right-skewed: R_{total} has mean 17.0%, median 13.1%, with a [P5,P95] band of [4.0%, 43.1%] – i.e. even at the 95th percentile of favourable pre-analytical conditions, at most ∼43% of the wound-fluid enzyme reaches the chamber.

Sensitivity. A binned first-order (Sobol-style) sensitivity decomposition on the same Monte Carlo sample (Fig. 3, left) attributes 83% of the variance in $[E]_{chamber}$ to V_{abs} alone, with D_f (3%) and η_{elut} (2%) a distant second and third, and $t_{transit}$ and k_{deg} each below 1% at these parameter ranges. This confirms contents.md Sec. 1.7's qualitative call that swab absorption volume dominates the pre-analytical uncertainty budget, and quantifies it: V_{abs} should be the first target of a dedicated wet-lab measurement campaign (swab weighing/gravimetric absorption studies across exudate levels and swab materials), well ahead of refining the elution-efficiency or degradation-rate placeholders. This single-layer ranking is a preview of the full cross-layer Sobol analysis planned for Layer 16.

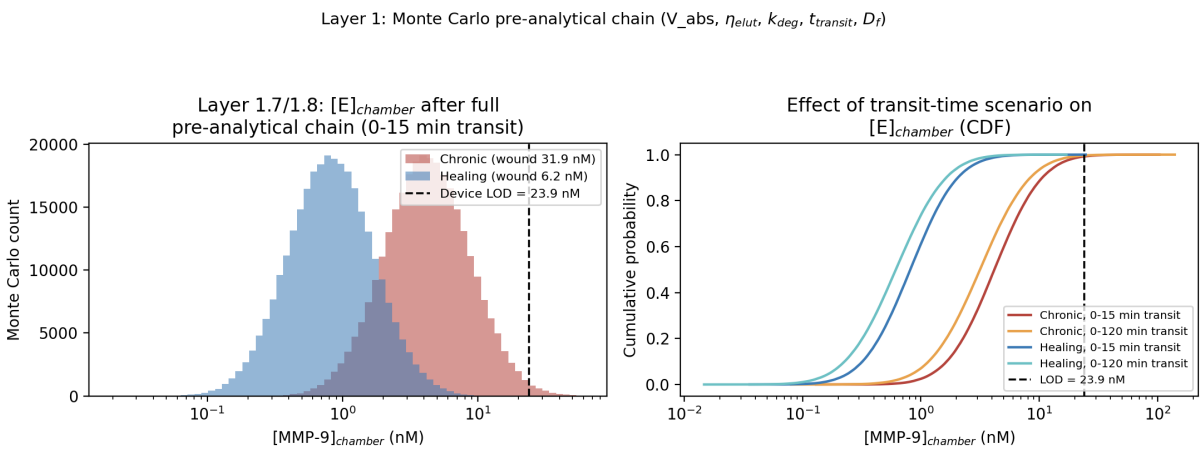


Figure 2: **Layer 1.7/1.8: Monte Carlo** $[E]_{chamber}$**.** Left: histograms of $[E]_{chamber}$ for the chronic and healing wound states (0–15 min transit), against Layer 15's LOD (23.9 nM, dashed). Right: cumulative distributions for both wound states under the clinic (0–15 min) and field (0–120 min) transit-time scenarios.

IX. Cross-Layer Integration Check

Layer 0's own text states that its wound-fluid outputs $[\text{MMP-9}]_{free}(t)$ and $[\text{HNE}]_{free}(t)$ "pass to Layer 1 (sample collection) via a swab dilution factor D_f" before reaching the device. In practice, Layer 9 (Sec. II, Table 1) takes $[E]_t = [\text{MMP-9}]_{free}$ *directly* from the Layer 0 ODE output – 30 nM chronic, 6 nM healing – and states explicitly that "TIMP-1 inhibition is fully accounted for in these values; no further correction is needed here." No swab dilution, elution efficiency, transit decay, or chamber-loading correction is applied anywhere in Layer 9, 13 or 15's chain: effectively, $R_{total} \equiv 1$ is assumed by omission. This report's own central-value estimate is $R_{total} = 7$–13% (Sec. VII), with a Monte Carlo 90% interval of roughly 4–43% (Sec. VIII) – i.e. $R_{total} = 1$ sits far outside even the optimistic tail of the propagated uncertainty.

Overlaying this report's $[E]_{chamber}$ distribution on Layer 15's own LOD (23.9 nM at the synthetic placeholder k_{surf}, Layer 15 Sec. III) makes the consequence concrete: Table 1 shows $P([E]_{chamber} < \text{LOD}) \geq 99\%$ for *both* the chronic and healing states under either transit scenario. Layer 15's own cross-layer check (its Sec. VII) already found the classifier's decision boundary (13.4 nM) falls below the LOD – a problem localised to concentrations *near* the clinical boundary. Folding in Layer 1's pre-analytical dilution suggests the more severe possibility that, unless the fabricated device's k_{surf} substantially exceeds the current placeholder *or* the swab/elution protocol is re-engineered to raise R_{total} (larger V_{abs}/smaller V_{buffer}, or forgoing dilution into buffer entirely in favour of a smaller elution volume), the device may be blind across the *entire* clinical concentration range, not only near the discrimination boundary.

Two reconciliations are possible and neither can be resolved from the documents reviewed for this report: (a) Layer 9/13/15's "clinical range (6–30 nM)" is implicitly intended as an already-diluted, chamber-level quantity (in which case Layer 0's wound-fluid values are actually higher, and Layer 0's own text describing a swab dilution step to reach "device range" supports this reading), or (b) the wound-fluid values are meant to be used directly as stated, and this layer's dilution chain has not yet been integrated downstream. Given Layer 0's figure caption explicitly reads "Layer 1 applies swab

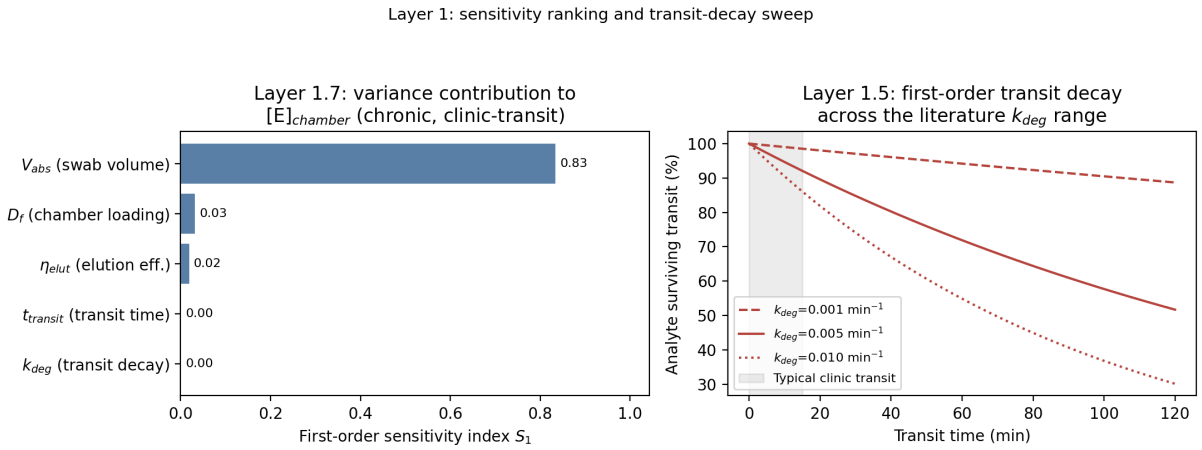


Figure 3: **Layer 1.7/1.5: sensitivity ranking and transit decay.** Left: first-order (binned Sobol) sensitivity indices for $[E]_{chamber}$ (chronic wound state, clinic-transit scenario). Right: first-order transit-decay curves (Eq. V.1) across the literature k_{deg} range, with the typical clinic transit window shaded.

dilution factor D_f to reach device range," interpretation (a) is more consistent with the existing figures, but this requires an explicit statement (and a defined D_f, η_{elut}, V_{abs} triple) in Layer 0 or Layer 9's own report to remove the ambiguity. This is the single highest-priority integration item raised by this report (Limitations, item 3).

X. Key Outputs

Output	Value (this report)	Feeds
Corrected eluate formula (Eq. II.1)	adds V_{abs}/V_{buffer} term	Layer 3, supersedes 1.2
V_{abs} distribution	LogNormal, GM 63.2 μL, GSD 2.01×	Sec. VIII, Layer 16 Sobol
Composite recovery R_{total} (15-min transit)	mean 17.0%, median 13.1%	Layer 3 input scaling
$[E]_{chamber}$, chronic (15-min transit)	median 4.18 nM [1.26, 13.81]	Layer 3, 9
$[E]_{chamber}$, healing (15-min transit)	median 0.81 nM [0.25, 2.67]	Layer 3, 9
$P([E]_{chamber} < \text{LOD})$, both states	≥99%	Sec. IX priority finding
Dominant uncertainty source	V_{abs}, $S_1 = 0.83$	Wet-lab priority list
SLPI/HNE ratio enrichment in chamber	1.45×	Layer 3 in-device inhibition
$\text{pH}_{chamber}$ (worked example)	7.02 (vs. 7.4 assumed)	Layers 6, 10, 12
Filter pressure drop (illustrative)	0.60 kPa	Layer 2 fill-pressure budget

XI. Limitations

1. V_{abs} **is not decomposed by exudate level or swab material.** contents.md Sec. 1.1 gives one pooled range (20–200 μL, "moderate exudate"); low- and heavy-exudate wounds, and different swab materials, are expected to shift this substantially. Since V_{abs} drives 83% of the pre-analytical variance (Sec. VIII), this is the single highest-value wet-lab measurement for the whole layer.

2. **Elution efficiency is MW-banded only, with no swab-material term, and no source**

is confirmed for the fabricated device's swab. Sec. II's caution (Wiegand & Hipler 2013 [5]; Singh et al. 2022 [6]) shows material effects can drive $\eta_{elut} \to 0$ for a chemically similar (superabsorbent) material; the cotton/foam/polyester materials contents.md specifies have not been tested for MMP-9/HNE elution specifically.

3. **The Layer 0→Layer 9 hand-off does not visibly apply this layer's dilution chain (Sec. IX).** Whether the "clinical range (6–30 nM)" used throughout Layers 9/13/15 is a wound-fluid or a chamber-level quantity is ambiguous across the existing reports and needs an explicit reconciliation before Layer 15's LOD/classifier numbers can be trusted at face value.

4. k_{deg} **(auto-proteolysis / transit degradation rate) is a bare literature range (0.001–0.01 min^{-1}) with no MMP-9/HNE-in-wound-fluid-eluate-specific source confirmed;** contents.md itself flags this as something to "test internally if not in literature."

5. **The filter pressure drop (Sec. VI) uses illustrative flow-rate and geometry inputs**, not Layer 2's actual chamber fill-time and capillary-pressure output (Layer 2 has not yet been modelled in this pipeline); the conclusion that filtration does not stall fill is provisional pending that cross-check.

6. **The D_f Monte Carlo spread (±15% relative) is an assumed pipetting/loading tolerance**, not sourced from contents.md, which gives only the single worked value $D_f = 0.8$.

7. **Inhibitor kinetics inside the chamber (Layer 3) are not modelled here.** Section III computes only the co-elution ratio shift; the actual competitive-inhibition dynamics during the 30-min incubation are Layer 3's responsibility, using this layer's $[I]_{chamber}$ values as initial conditions.

References

Papers confirmed in Layers 0 and 9's own reference lists are drawn on directly where noted; two new citations are added in this report to support Sec. II's elution-efficiency material-dependence caution.

1. Ladwig GP, Robson MC, Liu R, Kuhn MA, Muir DF, Schultz GS (2002). Ratios of activated matrix metalloproteinase-9 to tissue inhibitor of metalloproteinase-1 in wound fluids are inversely correlated with healing of pressure ulcers. *Wound Repair Regen* 10(1):26–37. – chronic/healing active MMP-9 concentrations confirmed (Layer 0; used directly as $[E]_{wound}$ in Sec. VII of this report).

2. Kridel SJ, Chen E, Kotra LP, Howard EW, Mobashery S, Smith JW (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276(23):20572–20578. DOI: 10.1074/jbc.M100900200. – MMP-9 molecular weight and substrate kinetics context confirmed (Layer 9).

3. Trengove NJ, Stacey MC, Macauley S, Bennett N, Gibson J, Burslem F, Murphy G, Schultz G (1999). Analysis of the acute and chronic wound environments: the role of proteases and their inhibitors. *Wound Repair Regen* 7(6):442–452. – HNE and TIMP-1 wound-fluid concentrations confirmed (Layer 0).

4. Rao CN, Ladin DA, Liu YY, Chilukuri K, Dong Z, Bhagwat SV (1995). α_1-Antitrypsin is degraded and non-functional in chronic wound fluid but intact and functional in acute wound fluid. *J Invest Dermatol* 105(4):572–578. – A1AT degradation in chronic wound fluid confirmed (Layer 0); relevant to A1AT co-elution behaviour, Sec. III.

5. Wiegand C, Hipler U-C (2013). A superabsorbent polymer-containing wound dressing efficiently sequesters MMPs and inhibits collagenase activity in vitro. *J Mater Sci Mater Med* 24(10):2473–2478. DOI: 10.1007/s10856-013-4990-6. – superabsorbent polyacrylate dressing shown to bind MMP-2/MMP-9 so strongly that a PBS elution step released none/marginal amounts; confirmed and newly cited here (Sec. II) as the basis for the swab-material elution-efficiency caution.

6. Singh G, Byrne C, Thomason H, McBain AJ (2022). Investigating the microbial and metalloprotease sequestration properties of superabsorbent wound dressings. *Sci Rep* 12:4747. DOI: 10.1038/s41598-022-08361-3. – most commercial superabsorbent dressings tested fully sequester MMP-2/MMP-9 over multi-day contact; confirmed and newly cited here (Sec. II) as corroborating evidence for the same caution.